

BANK LOAN ANALYSIS DASHBOARD

1. Project Overview

The Bank Loan Analysis Dashboard is an interactive Business Intelligence solution developed using Power BI to analyze loan performance, customer behavior, and lending trends. The dashboard enables stakeholders to monitor loan applications, funded amounts, repayments, loan status distribution, and borrower characteristics through dynamic visualizations and KPIs. The project helps financial institutions evaluate loan portfolio performance, identify good and bad loans, and make data-driven lending decisions.

2. Business Problem

Banks process thousands of loan applications and require a comprehensive reporting system to:

- Monitor loan application trends.
- Track funded and received amounts.
- Identify good and bad loans.
- Analyze customer demographics.
- Evaluate repayment performance.
- Support strategic lending decisions.

Traditional reporting methods are time-consuming and do not provide real-time insights. This dashboard solves these challenges by presenting data in an interactive and visually appealing format.

3. Project Objectives

The primary objectives of this project are:

- Analyze loan portfolio performance.
- Measure loan funding and repayment trends.
- Identify Good Loans and Bad Loans.
- Track loan applications across states.
- Analyze loan purpose and borrower characteristics.
- Provide executives with a high-level summary.
- Enable interactive filtering and drill-down analysis.

4. Tools and Technologies Used

Tool	Purpose
Power BI Desktop	Dashboard Development
Power Query	Data Cleaning and Transformation
DAX	KPI and Measure Creation

Excel / CSV Dataset	Source Data
Data Modeling	Relationship Management

5. Data Cleaning and Transformation

The following preprocessing steps were performed:

Date Transformation

- Converted Issue Date into Date format.
- Created Month and Quarter columns.

Data Type Corrections

- Interest Rate → Decimal Number
- DTI → Decimal Number
- Funded Amount → Currency
- Loan Amount → Currency

Handling Missing Values

- Checked null values.
- Removed unnecessary blanks.

Feature Engineering

Created:

- Issue Month
- Quarter
- Good Loan Flag
- Bad Loan Flag

6. DAX Measures Created

Loan Application Measures

Total Loan Applications =COUNT(LoanData[id])

Funded Amount

Total Funded Amount =SUM(LoanData[loan_amount])

Amount Received

Total Amount Received =SUM(LoanData[total_payment])

Average Interest Rate

Average Interest Rate =AVERAGE(LoanData[int_rate])

Average DTI

Average DTI =AVERAGE(LoanData[dti])

Good Loan Applications

Good Loan Applications =CALCULATE(COUNT(LoanData[id]), LoanData[loan_status] IN {"FullyPaid","Current"})

Good Loan Percentage

Good Loan % =DIVIDE([Good Loan Applications],[Total Loan Applications])

Bad Loan Applications

Bad Loan Applications =CALCULATE(COUNT(LoanData[id]),LoanData[loan_status] = "ChargedOff")

Bad Loan Percentage

Bad Loan % =DIVIDE([Bad Loan Applications], [Total Loan Applications])

7. Dashboard Pages

Page 1: Executive Summary

KPIs

- Total Applications
- Total Funded Amount
- Total Amount Received
- Average Interest Rate
- Average DTI

Visualizations

- Monthly Application Trend
- Loan Status Distribution
- State-wise Good Loan Analysis
- State-wise Bad Loan Analysis

Purpose

Provides an overall performance overview of the loan portfolio.

Page 2: Good Loan Analysis

KPIs

- Good Loan Applications
- Good Loan Funded Amount
- Good Loan Amount Received
- Good Loan Percentage

Visualizations

- Grade Analysis
- Quarterly Trend Analysis
- State Analysis
- Purpose Analysis
- Home Ownership Analysis

Purpose

Identifies characteristics of successfully performing loans.

Page 3: Bad Loan Analysis

KPIs

- Bad Loan Applications

- Bad Loan Funded Amount
- Bad Loan Amount Received
- Bad Loan Percentage

Visualizations

- Grade Analysis
- Quarterly Trend Analysis
- State Analysis
- Purpose Analysis
- Home Ownership Analysis

Purpose

Analyzes risky and defaulted loans to reduce future losses.

Page 4: Business Analysis

Visualizations

1. Monthly Trend Analysis
2. Regional Analysis
3. Loan Term Analysis
4. Employment Length Analysis
5. Loan Purpose Analysis
6. Home Ownership Analysis

Purpose

Provides deep insights into borrower behavior and lending performance.

8. Key Findings

Loan Portfolio Performance

- Total Applications: 39K
- Total Funded Amount: 436M
- Total Amount Received: 473M

Good Loans

- Approximately 86% of loans are classified as Good Loans.
- Majority of good loans belong to Grades A and B.

Bad Loans

- Approximately 14% of loans are classified as Bad Loans.
- Higher risk observed in lower credit grades.

Loan Purpose

- Debt Consolidation is the most common loan purpose.
- Credit Card and Home Improvement are significant categories.

Home Ownership

- Most borrowers belong to Mortgage and Rent categories.

Regional Trends

- Certain states contribute significantly more loan applications than others.

9. Business Impact

The dashboard helps stakeholders:

- Monitor lending performance.
- Reduce loan default risk.
- Improve loan approval strategies.
- Understand customer borrowing patterns.
- Enhance portfolio management.
- Support strategic decision-making.

10. Conclusion

The Bank Loan Analysis Dashboard successfully transforms raw loan data into meaningful business insights through interactive visualizations, KPIs, and analytical reports. The solution enables financial institutions to evaluate portfolio health, identify risk factors, and optimize lending decisions. The dashboard provides a complete view of loan performance through Executive Summary, Good Loan Analysis, Bad Loan Analysis, and Business Analysis pages, making it a valuable tool for business users and decision-makers

11.Screenshots.

